| Research question | Statistical approach | R function |
|---|---|---|
| Do the 3 stations differ in RQA metrics? (Unpaired distribution comparison) | One-Way ANOVA (parametric) and Kruskal-Wallis + BH-adjusted Wilcoxon (non-parametric) | `aov(), kruskal.test(), pairwise.wilcox.test()` |
| Do the 3 stations differ when matched by DOY? (Paired by day of year) | Non-parametric Friedman Rank Sum Test + Kendall's $W$ | `friedman.test()` |
| Do the 3 stations differ across all 8 metrics jointly? (Multivariate profile) | Multivariate Analysis of Variance (MANOVA, Pillai's trace) | `manova()` |
| Does a station match a synthetic model on individual metrics? (Sample vs reference value) | One-Sample $t$-test and One-Sample Wilcoxon Signed-Rank Test against $\mu =$ model value | `t.test(x, mu = val), wilcox.test(x, mu = val)` |
| Which dynamical regime best describes each station? (Multi-metric classification) | Mahalanobis Distance and Standardised Euclidean ($z$-score) Distance + PCA Biplot | `mahalanobis(), prcomp()` |